DSA 210: Term Project Final Report

Impact of Fuel Prices and Holidays on Flight Ticket Pricing: A Machine Learning Approach

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Course: Introduction to Data Science

1. Introduction & Motivation

The aviation industry is characterized by highly dynamic and complex pricing models. As a student frequently looking for affordable travel options, the seemingly random fluctuations in flight ticket prices served as the primary motivation for this project.

While it is well documented that internal flight characteristics—such as flying Business Class or booking a long-haul 10-hour flight—drastically increase ticket costs, the impact of external macro-environmental factors is less obvious to the average consumer. Therefore, the core objective of this term project was to rigorously investigate two external variables: **Global Crude Oil Prices (WTI)** and **National Holidays**.

Initial Hypotheses:

1. *Fuel Cost Dependency:* Since aviation fuel is a massive operational cost for airlines, a spike in global crude oil prices should theoretically trigger an immediate increase in consumer ticket prices.
2. *Holiday Demand Surge:* National public holidays generate massive spikes in travel demand. Basic supply-and-demand economics suggests that flight tickets should be significantly more expensive on these dates.

2. Data Collection, Cleaning, and Enrichment

To conduct this analysis, a singular dataset was insufficient. A rigorous data enrichment process was performed using the **pandas** and **numpy** libraries.

2.1 Primary Dataset: The foundational data was sourced from Kaggle's 'Flight Price Prediction' dataset, encompassing roughly 300,000 flight records across India. The raw data included columns such as `airline`, `flight`, `source_city`, `departure_time`, `stops`, `arrival_time`, `destination_city`, `class`, `duration`, `days_left`, and `price`.

2.2 Data Preprocessing & Cleaning: Several critical transformations were executed in the Jupyter Notebooks:

- *Duration Parsing:* The `duration` column initially contained string values (e.g., '2h 50m'). A custom Python function was written to extract the hours and minutes using string splitting, converting the entire column into a continuous numeric `duration_minutes` feature.
- *Dropping Irrelevant Features:* The `flight` code column (e.g., 'UK-706') was dropped entirely as it had too high a cardinality and provided no generalized predictive value for the model.

2.3 Data Enrichment via Merging:

• *Oil Prices:* Historical WTI Crude Oil daily spot prices were scraped from Investing.com. Because oil markets are closed on weekends, missing values (`NaN`) were generated when merging with the flight dataset. I utilized the `ffill` (forward fill) interpolation method in pandas to carry Friday's oil price over the weekend, ensuring every flight record had an associated fuel cost.

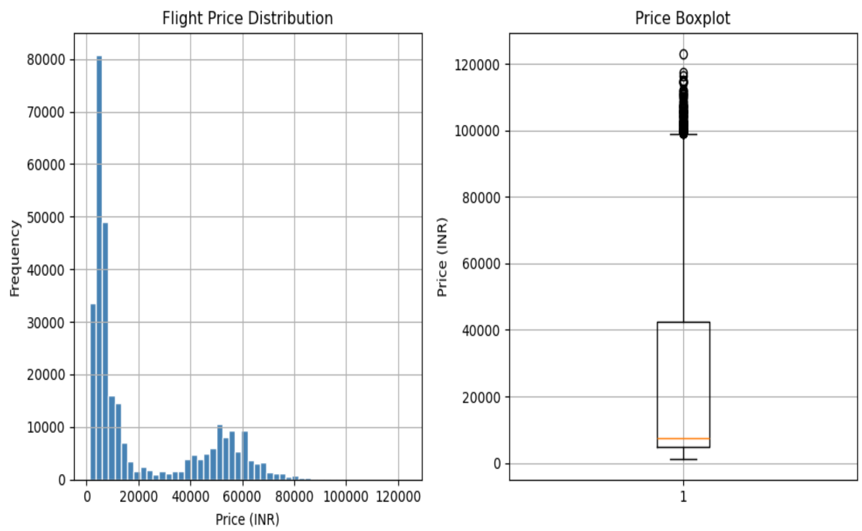
• *Holidays:* A worldwide holiday dataset was filtered down strictly to relevant national holidays. I engineered a binary `is_holiday` feature mapping to the flight dates using a left `pd.merge`.

3. Exploratory Data Analysis (EDA) & Hypothesis Testing

Before building predictive models, exploratory data analysis was conducted using **matplotlib** and **seaborn** to understand the data distributions and test the initial hypotheses.

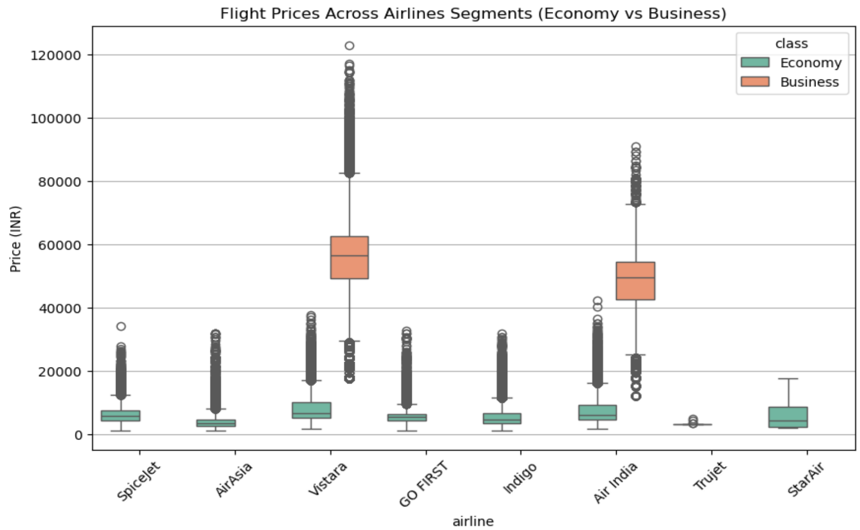
3.1 Understanding the Baseline: Ticket Price Distribution

A histogram visualization of the target variable (`price`) revealed a severe right-skew. The vast majority of ticket prices are clustered in the lower range (Economy class), with a long tail stretching into the upper brackets representing premium seats and last-minute bookings. This skew indicated that a linear model might struggle with extreme outliers.



3.2 Internal Factors: The Class Divide

To baseline the internal factors, I plotted the average prices across different airlines segmented by Cabin Class. The visualization starkly demonstrated that 'Business' class tickets are exponentially more expensive than 'Economy', regardless of the airline. Vistara and Air India completely dominated the premium market space.

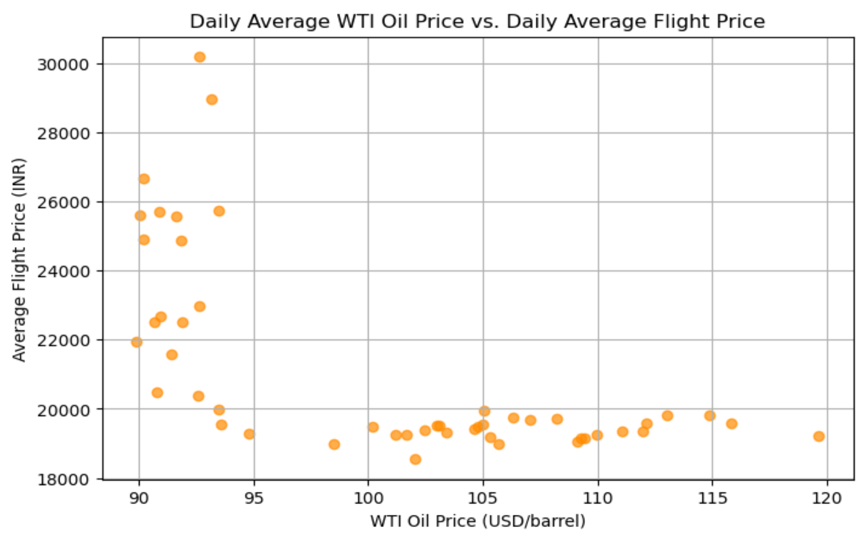


Flight Prices Across Airlines Segments (Economy vs Business)
class
Economy
Business
Price (INR)
0
20000
40000
60000
80000
100000
120000
SpiceJet
AirAsia
Vistara
GO FIRST
Indigo
Air India
Trujet
StarAir
airline

3.3 Testing Hypothesis 1: The Oil Price Paradox

To test if higher oil prices correlate with higher ticket prices, I grouped the data by date and calculated the daily averages. I utilized `scipy.stats.pearsonr` to compute the Pearson correlation coefficient.

Result: The correlation was calculated at *r = -0.6519*. The negative sign meant the hypothesis was entirely wrong—as oil prices went up in my timeframe, ticket prices actually trended downwards. Recognizing that airlines purchase fuel via 'hedging' contracts months in advance, I simulated 7-day and 14-day lags. The correlation weakened (r = -0.49 for 7-day), proving that daily spot oil prices do not immediately dictate consumer ticket costs.



3.4 Testing Hypothesis 2: Holidays are Cheaper?!

To evaluate the holiday demand surge, I separated the dataset into 'Holiday' and 'Non-Holiday' groups and ran a Welch's t-test using `scipy.stats.ttest_ind(equal_var=False)`. The p-value was effectively 0.0, indicating statistical significance.

Result: Astonishingly, the mean holiday ticket price was 19,506 INR, while normal days averaged 20,946 INR. Flights were statistically *cheaper* on holidays. The working theory is a demographic shift: highly lucrative corporate/business travel ceases during public holidays, and the influx of cheaper economy leisure travel drags the overall mathematical mean downwards.

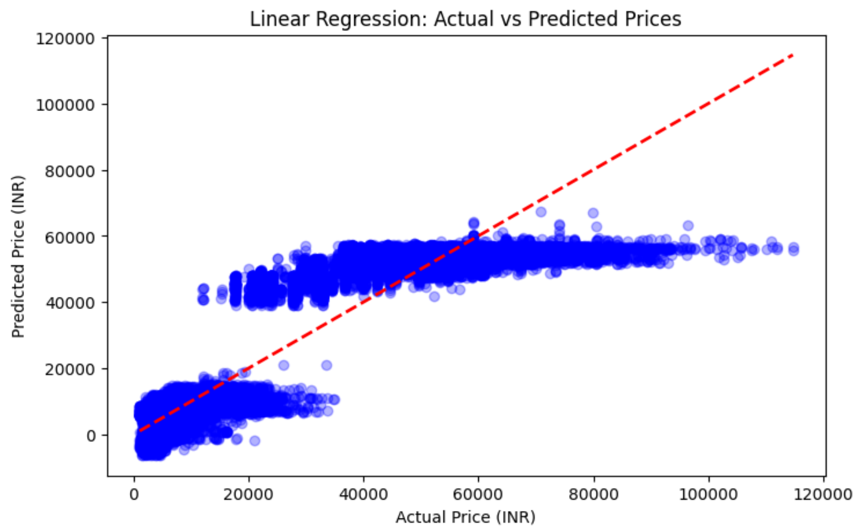
Flight Prices: Holiday vs. Non-Holiday
Price (INR)
120000
100000
80000
60000
40000
20000
0
Holiday
Non-Holiday

4. Machine Learning & Predictive Modeling

In the final notebook phase, the goal was to build predictive models using **scikit-learn**. The dataset was prepared by applying `pd.get_dummies` to one-hot encode all categorical variables (`airline`, `source_city`, `destination_city`, `departure_time`, `arrival_time`, `class`). Numerical features (`duration_minutes`, `days_left`, `oil_price`) were scaled using `StandardScaler` to prevent larger magnitudes from dominating the loss functions. An 80-20 Train-Test split was utilized.

4.1 The Baseline: Linear Regression

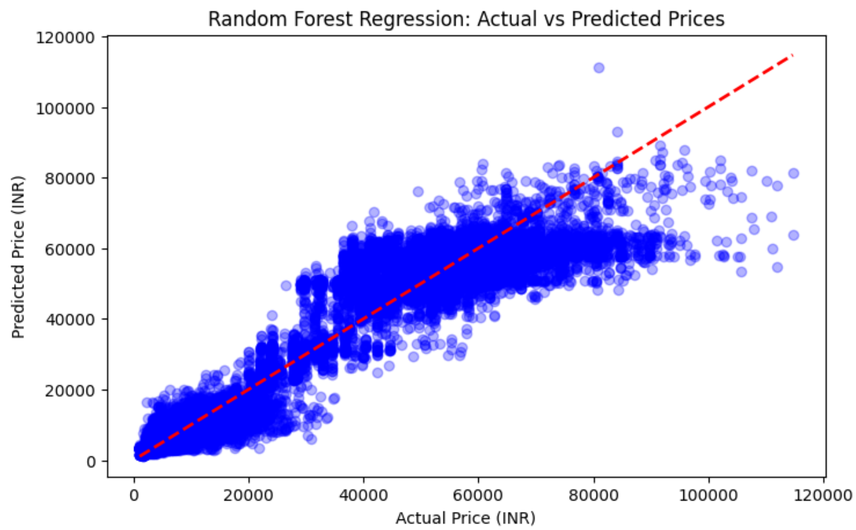
A standard Linear Regression model was fitted first. It achieved a very respectable R-squared score of 0.9049 and a Root Mean Squared Error (RMSE) of 6,995 INR. However, as seen in the scatter plot below, the straight-line prediction severely struggles with higher-priced (Business class) tickets, forming distinct clusters rather than a cohesive line.



4.2 The Heavyweight: Random Forest Regressor

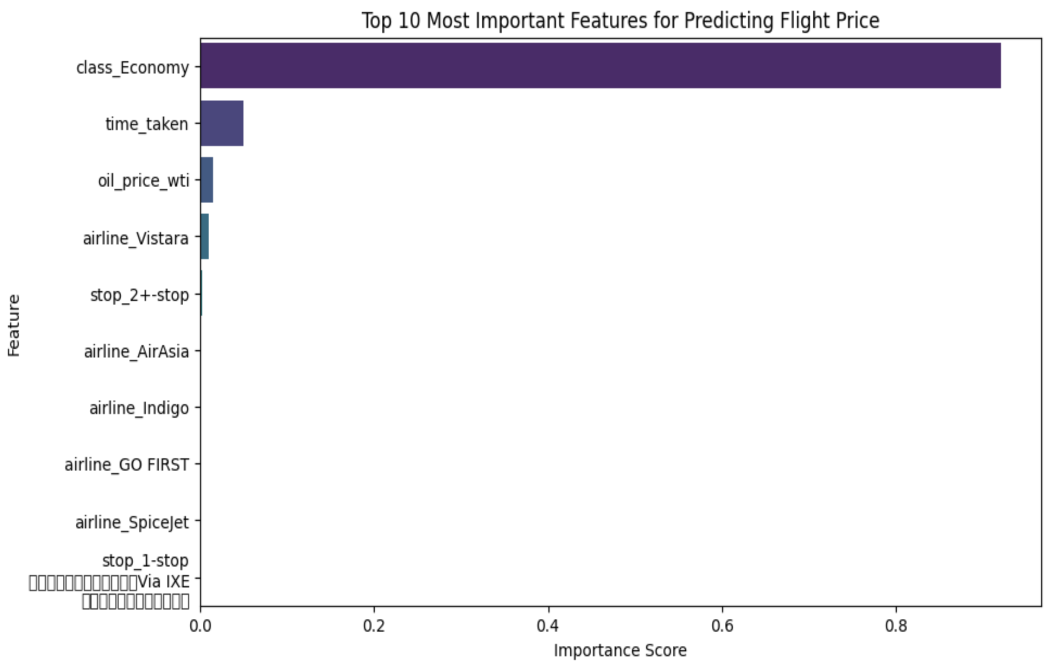
To capture the non-linear tier pricing of airline tickets, an ensemble method was deployed. Given the 300,000 row dataset, running a full Random Forest is computationally expensive. Therefore, I utilized `GridSearchCV` combined with K-Fold Cross Validation to systematically test hyperparameters. The optimal configuration was found to be `n_estimators=100` and `max_depth=15`.

The performance improvement was massive: the R-squared score jumped to 0.9493, and the RMSE dropped significantly to 5,110 INR. The scatter plot shows a much tighter, more accurate clustering along the identity line.


Random Forest Regression: Actual vs Predicted Prices
Predicted Price (INR)
Actual Price (INR)
0
20000
40000
60000
80000
100000
120000

5. Conclusion: What Really Matters?

To finally address the core project question, I extracted the `feature_importances_` attribute from the optimally trained Random Forest model. This metric calculates how much each feature contributes to decreasing the model's impurity (MSE) during training.



The visual evidence is indisputable. The model proved that internal flight characteristics absolutely dominate pricing. The binary indicator for `class_Economy` / `class_Business` is by far the most critical predictor, followed by flight duration and the number of days left before departure.

Final Takeaway: While macro-environmental variables like global oil markets and national holidays make for interesting hypotheses, their actual predictive power in the algorithm is virtually zero when compared to basic flight logistics. Airlines price tickets based fundamentally on the seat tier chosen and the distance flown. The data science journey conclusively disproved the initial hypotheses while simultaneously producing a highly accurate, 95% reliable predictive engine.